CIVICVOICE

A Community Issue Reporting and Management Platform

Dissertation Planning Document

Proposed structure, scope, and requirements — for approval

This is a planning document, not the dissertation. No dissertation chapter content has been written yet. It records the study of the reference BCA dissertation, the verified state of the CivicVoice source code, the proposed table of contents and chapter structure, the information required to write accurately, and the diagrams, tables, and screenshots to be produced.

Document control	Detail
Project	CivicVoice — community issue reporting and management platform
Document type	Dissertation planning document (pre-writing)
Reference dissertation	Mentor Lab: An AI-Powered Educational Platform for BCA, BBA & Biotechnology Students (P.K. Roy Memorial College, Dhanbad, September 2026)
Source inspected	C:\Users\atulp_5y4afxd\civicvoice-mine (branch main, 33 commits, latest 2026-10-23)
Date	23 October 2026
Status	Awaiting approval — five scope decisions required

1. Reference document studied

I located and studied the reference dissertation:

- **File:** C:\Users\atulp_5y4afxd\Downloads\pk roy doc.docx
- **Title:** "MENTOR LAB: AN AI-POWERED EDUCATIONAL PLATFORM FOR BCA, BBA & BIOTECHNOLOGY STUDENTS"
- **Programme:** BCA, P.K. Roy Memorial College, Dhanbad, September 2026.

I extracted its complete text and its Word formatting XML. There is also "Adobe Scan 23 Oct 2026.pdf" (21 pages, image-only, no text layer). I cannot read image-only PDFs in this session, so I used the DOCX. If the scanned PDF is the intended reference, please provide an OCR'd/text version or confirm the DOCX is correct.

1.1 What I learned from the reference

Aspect	Reference pattern
Overall structure	Preliminary pages → 9 chapters → References → Appendix (Source Code)

Aspect	Reference pattern
Preliminary pages	Title page, Certificate, Certificate of Approval, Acknowledgment, Abstract, Index, List of Figures, List of Tables
Chapter pattern	“CHAPTER N” + title, numbered sections/subsections, each chapter ends with “N.x Summary” that previews the next chapter
Chapter sequence	1 Introduction, 2 Literature Review, 3 Methodology, 4 Analysis, 5 System Design, 6 Implementation, 7 Testing and Performance Evaluation, 8 Application, 9 Conclusion
Writing style	Third-person academic; past tense for work performed, present tense for system behaviour; short paragraphs (1–3 per subsection); bullet lists for objectives/features
Formatting (from DOCX XML)	A4; Times New Roman 12 pt body; 14 pt chapter headings; 13 pt section headings; 9 pt captions; Courier New 9 pt for code; justified text; exact 18 pt line spacing; margins 1″ top/right/bottom, 1.25″ left (binding); header/footer 0.5″
Numbering	“Fig. X.Y” below figures; “Table X.Y” above tables; numbered references
Level of detail	~44 pages main body + ~16-page source-code appendix; 15 figures; 10 tables

2. Critical discrepancies to resolve before writing

Your brief and the code in civicvoice-mine do not currently match on five points. I will not silently choose one — these affect the table of contents, modules, ER diagram, API tables, test cases, and appendix.

#	Your brief	Actual code in civicvoice-mine	Proposed handling
1	Admin Panel removed	Still present: frontend/src/pages/AdminDashboard.jsx, /admin route in App.jsx:64, backend/src/routes/admin.js (5 endpoints), admin role in User.js, scripts/seedAdmin.js	Follow your instruction and exclude admin from the dissertation. But the appendix/source code must match the submitted build — please provide the cleaned version, or confirm I should omit those files from the appendix while leaving the repository unchanged.
2	Analytics Dashboard removed	No standalone dashboard; only an "Analytics" tab inside AdminDashboard.jsx:89	Excluded together with the admin panel.
3	Workflow: Submitted → Verified → Prioritized → In Progress → Resolved	Issue status enum is pending, in-progress, resolved, closed; verified and prioritized do not exist as statuses	Either (a) document the implemented 4-stage flow, or (b) document your 5-stage flow as a conceptual model with a mapping table (Submitted = pending, Verified = official review, Prioritized = aiPriority, In Progress = in-progress, Resolved = resolved, Closed = post-resolution). Please choose.
4	Stack includes Framer Motion and React Hot Toast	Neither is in either package.json; the app uses a custom Toast.jsx component and CSS transitions	Document the actual implementation, or send the build that uses these libraries. I will not claim them if they are absent.
5	Roles not specified	resident, official, admin	Document Resident and Official; mention admin only if the submitted build retains the role.

Also noted: PWA support is implemented (vite-plugin-pwa, service worker, install prompt) even though it is not in your feature list. I propose including it as an implemented feature unless you say otherwise.

3. My understanding of CivicVoice

CivicVoice is a web-based community issue reporting and management platform. A citizen registers, reports a local problem (damaged road, garbage, water supply, drainage, streetlight, etc.) with a description, category, map location, and up to five photographs, and the issue becomes visible to the community. Other users can vote and comment. Google Gemini analyses the report and returns an AI priority score (1–10), 2–3 practical suggestions, and a step-by-step solution. Issues are displayed on an OpenStreetMap-based interactive map with colour-coded, clustered markers. Officials update the issue status and attach progress photos; the reporter receives in-app notifications. A Gemini-powered chatbot ("CivicBot") answers platform and issue-related questions.

The application is a three-tier system:

- **Presentation:** React 18 single-page application built with Vite 6, React Router 7, Tailwind CSS 4, React Leaflet, Axios, and a custom toast/theme/splash system.
- **Application logic:** Node.js + Express 4 REST API with JWT authentication, bcrypt password hashing, express-validator, express-rate-limit, Multer uploads, and Nodemailer OTP email.
- **Data:** MongoDB with Mongoose 8 (GeoJSON 2dsphere index on issue location), plus optional ImageKit cloud image storage.

3.1 Verified implemented features (from source)

- Registration with first/middle/last name, mobile, Gmail-only email, state/district, and consent; 6-digit OTP email verification; strong-password policy (8+ characters, upper/lower/number/symbol); JWT sessions valid for 7 days.
- Forgot/reset password via OTP; change password with current password + OTP; profile update; avatar upload/remove.
- Issue creation with title (3–120 characters), description (10+ characters), six categories, validated GeoJSON point, state, and up to 5 images (5 MB each).
- Gemini analysis: priority 1–10, up to 3 suggestions, one solution; asynchronous after creation and after each vote; manual POST /api/issues/:id/analyze.
- Explore feed with category/status/state filters, four sort modes, and pagination; issue statistics endpoint.
- Interactive maps using OpenStreetMap tiles, React Leaflet, marker clustering, and Nominatim search/reverse geocoding.
- One-vote-per-user toggle, comments, and in-app notifications (status change, resolution, vote, comment).
- Status update log with per-update photos; status changes notify the author.
- Resident dashboard (/dashboard), notifications page, profile page.

- CivicBot chat endpoint (rate-limited, Gemini-backed, graceful fallback).
- PWA install support, dark/light theme, mobile bottom navigation, splash screen, contact form, and static information pages.

Verified database collections: User, Issue, Otp, Notification, Contact.

Verified API surface: 12 auth endpoints, 11 issue endpoints, 4 notification endpoints, 1 contact endpoint, 1 upload endpoint, 1 chat endpoint (plus 5 admin endpoints that fall outside your stated scope).

4. Complete proposed table of contents

Page numbers are provisional and will be finalised after typesetting. Planned length: ~48 pages main body + ~16-page appendix.

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Note: Chapter 8's "8.1 Summary" mirrors the reference exactly. If your department prefers, it can be renamed "8.1 Introduction"; otherwise I will keep the reference layout.

5. Chapter-wise structure

Chapter 1 — Introduction

- **1.1 Introduction:** the civic-reporting problem, fragmented complaint channels, lack of transparency and prioritisation; introduce CivicVoice as a geotagged, photo-backed, community-voted, AI-assisted platform.
- **1.2 Motivation:** civic participation, accountability, hyperlocal relevance, and the practical value of AI-assisted triage. I will not invent specific incidents or survey statistics.
- **1.3 Aims and Objective:** 7–8 bullets tied strictly to implemented features — reporting with geolocation and photos, voting/commenting, Gemini priority/suggestions/solution, OSM map with clustering, lifecycle tracking, notifications, secure OTP authentication, responsive/PWA access.
- **1.4 Summary:** chapter recap and roadmap.

Chapter 2 — Literature Review

- **2.1 Introduction:** purpose and scope of the review.
- **2.2 Background:** civic technology, e-government, crowdsourced issue reporting, GIS/mapping, and AI-assisted prioritisation.
- **2.3 Some Related Works:** 4–5 comparable platforms, each with strengths and limitations. Candidates to research and cite: FixMyStreet (UK), SeeClickFix (US), Swachhata-MoHUA (India), and municipal grievance/311 portals. All factual claims and citations will be verified before writing; I will not invent dates or features.
- **2.4 Summary:** the gap CivicVoice addresses (hyperlocal reporting + OSM + AI prioritisation + community voting in one open, India-oriented platform).

Chapter 3 — Methodology

- **3.1 Introduction:** why a structured methodology was needed.
- **3.2 SDLC:** definition and phases, then each sub-phase as applied to CivicVoice — feasibility, requirement study, analysis, design, coding, testing, implementation. Technology feasibility will cite the actual stack.
- **3.3 Agile Development Methodology:** iterative increments and Git-based version control. The repository has 33 commits from 9 August to 23 October 2026, which I can cite as evidence of iteration. I will not claim formal sprint ceremonies, stand-ups, or team roles unless you confirm them.
- **3.4 Summary:** chapter recap.

Chapter 4 — Analysis

- **4.1 Introduction:** purpose of system analysis.
- **4.2 Requirement Engineering:** definition and sources (observation, peer use, supervisor consultation, comparison with existing platforms). Placeholder for any survey/interview data you may have.
- **4.3 Requirement Engineering Process:** technical/operational/economic feasibility; gathering; SRS; validation.
- **4.3.3 SRS:** functional requirements as bullets — registration/OTP, reporting, AI analysis, voting, comments, map, lifecycle, notifications, profile, chatbot, PWA.
- **4.4 Software Requirement for Development:** Node.js 18+, npm, MongoDB, Gemini API key, VS Code, Git/GitHub, modern browser.
- **4.5 Hardware Requirement For Use:** table.
- **4.6 Software Requirement for Use:** table.
- **4.7 Use Case:** actors — Resident, Official, Visitor; Gemini and OSM treated as external services. Steps for designing use cases.
- **4.8 Use case Diagram:** three diagrams (Resident, Official, Visitor). Admin is excluded per your instruction.
- **4.9 Block Diagram:** client → REST API → MongoDB, with Gemini, OSM/Nominatim, SMTP, and optional ImageKit.
- **4.10 Flowchart:** report-an-issue flow and issue-lifecycle flow.
- **4.11 Summary:** chapter recap.

Chapter 5 — System Design

- **5.1 Introduction:** purpose of system design.
- **5.2 System Design:** three-tier architecture; frontend/backend folder structure; deployment model (render.yaml, build.sh).
- **5.3 DFD:** physical vs logical; symbols; Level 0 context diagram; Level 1 processes (Authentication, Issue Reporting, Voting/Commenting, Status Management, Notifications, AI Analysis/Chatbot).
- **5.4 Data Design:** ER diagram for User, Issue, Otp, Notification, Contact; relationships; embedded votes, comments, statusUpdates; indexes (2dsphere, unique email, notification and OTP compound indexes).
- **5.5 Activity Diagram:** report issue, vote, official status update, AI analysis, chatbot.
- **5.6 Summary:** chapter recap.

Chapter 6 — Implementation

- **6.1 Introduction:** implementation overview.

- **6.2 Software Implementation:** frontend and backend stack tables with actual versions from package.json; project structure; API endpoint tables; security summary (JWT 7-day, bcrypt cost 10, OTP hashed with bcrypt, rate limits, validation, CORS, Multer restrictions).
- **6.3 User Interface Design:** page/component inventory, theme system, responsive/bottom navigation, splash, toast, PWA install.
- **6.4 Screenshots:** 14–16 real captures (see Section 7). I will not fabricate screenshots.
- **6.5 Summary:** chapter recap.

Chapter 7 — Testing and Performance Evaluation

- **7.1 Introduction:** purpose of testing.
- **7.2 Software Testing:** verification and validation definitions and how they were applied.
- **7.3 Basics:** black-box and white-box testing.
- **7.4 Types:** unit, integration, system, regression, and user acceptance testing.
- **7.5 Test Example:** test-case table (ID, description, input, expected, actual, status). Every "Actual/Status" entry must come from a real test run. Until you provide evidence, the table will be marked "To be executed" — not "Pass".
- **7.6 Summary:** only include performance claims (e.g., response times) if we actually measure them; otherwise keep the summary qualitative.

Chapter 8 — Application

- **8.1 Summary:** mirrors the reference layout.
- **8.2 About the Application:** what the final application does, how it is accessed, and its scope (excluding admin/analytics).
- **8.3 Features of the Software:** verified feature list — authentication/OTP, reporting, AI analysis, explore/filter/sort, map, voting/comments, lifecycle, notifications, dashboard, CivicBot, PWA, theme, contact.
- **8.4 Summary:** chapter recap.

Chapter 9 — Conclusion

- **9.1 Conclusion:** what was built, how it addresses the problem, and the significance of the AI prioritisation and community-voting combination.
- **9.2 Limitations:** accurate limitations only — Gmail-only registration; OTP depends on SMTP (console fallback in development); AI depends on Gemini availability/quota and is advisory; local-disk uploads by default; in-app notifications only; English-only UI; web/PWA only; no automated test suite in the repository; status model limited to the implemented states; admin/analytics out of scope.

- **9.3 Future Scope:** email/push notifications, full-text/advanced search, official assignment/routing, real-time updates, multi-language support, native mobile apps, cloud storage/CDN, automated testing/CI, accessibility improvements, government API integration.

References

Numbered list in the reference's style, using real sources only: React, Vite, Tailwind CSS, React Router, Leaflet, React Leaflet, OpenStreetMap, Express, MongoDB, Mongoose, Google Gemini API, Axios, JWT, bcryptjs, Multer, Nodemailer, express-validator, Pressman, Sommerville.

Appendix — Source Code

Selected, relevant excerpts (not the whole repository): server.js, app.js, Issue.js, User.js, auth middleware, issues.js (create/vote/status), gemini.js, App.jsx, api.js, CreateIssuePage.jsx, IssueDetailPage.jsx, Chatbot.jsx. Approximately 12–16 pages.

6. Technical information and project files required

6.1 Already available and verified

Item	Location / status
Full source code	C:\Users\atulp_5y4afxd\civicvoice-mine (branch main, 33 commits, latest 2026-10-23)
Root/frontend/backend package.json	Available; exact versions extracted
.env.example	Available; lists MongoDB, JWT, Gemini, CORS, ImageKit, SMTP, admin bootstrap variables
README.md, CivicVoice Documentation.md	Available as inputs (both currently describe admin features, so they cannot be copied verbatim if admin is excluded)
render.yaml, build.sh, vite.config.js	Available
Models, routes, middleware, utilities	Available and inspected
Git history	Available; usable as evidence of iterative development
Reference dissertation	Downloads\pk roy doc.docx
Existing project dump	project_files.txt (723 KB, includes node_modules; not needed)

6.2 Required from you — identity and institution

Please provide these as exact text; I will use "[PLACEHOLDER: ...]" until then.

1. Final project title (confirm: "CivicVoice: A Community Issue Reporting and Management Platform" or your preferred wording).
2. Student full name, registration number, university roll number.
3. Degree/course, semester/year, session.
4. College and department names, university affiliation, address.
5. Supervisor name and designation.
6. Principal and coordinator names (for the certificate pages).
7. Month and year of submission.
8. Individual or group project — if group, all names and registration numbers.
9. College logo file (PNG/JPG, high resolution) for the title page.

6.3 Required from you — scope decisions

10. Confirm the clean build that truly excludes the admin panel and analytics dashboard, or confirm that I should simply omit those files/endpoints from the dissertation while the repository remains unchanged.
11. Confirm the issue workflow: implemented 4-state (pending → in-progress → resolved → closed) or your 5-state conceptual flow (with a mapping table).

12. Confirm whether Framer Motion and React Hot Toast are actually used. If yes, send that version; if not, I will document the custom Toast.jsx and CSS transitions.
13. Confirm the roles to document: Resident and Official only, or also Admin.
14. Confirm whether to include PWA, dark/light theme, CivicBot, and email OTP as implemented features (all are present in the code).
15. Confirm the deployment URLs to cite (the README lists Render URLs; the frontend api.js defaults to https://civicvoice-znpn.onrender.com/api).

6.4 Required from you — evidence and assets

16. Screenshots of the running application (or permission and environment to run it locally). I will not use mockups or invented UI.
17. Test evidence: manual test log, Postman results, or a confirmation that I should present the test cases as planned without claiming they passed.
18. Sample data counts (users/issues) if you want platform statistics in the report; otherwise I will omit numbers.
19. Performance measurements if you want the "Performance Evaluation" part of Chapter 7 to contain real figures; otherwise it stays qualitative.
20. Any college dissertation guidelines (font, margins, page limit, certificate wording, declaration page) that override the reference format.

6.5 Required from you — output preferences

21. Output format: DOCX (recommended, closest to the reference), PDF, LaTeX, or Markdown.
22. Diagram format: draw.io/Visio images, PlantUML, or Mermaid (I can generate Mermaid/PlantUML source and embedded images).
23. Citation style: the reference uses a numbered list; confirm that is acceptable.

Do not send secrets. I only need to know whether Gemini, SMTP, ImageKit, and MongoDB are configured — not their values.

7. Proposed diagrams, tables, and screenshots

7.1 Figures

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7.3 Screenshots

The 15 screenshots listed as Figs. 6.1–6.15 are the minimum set. They must be captured from the actual application at a consistent resolution and browser width. I will add captions and annotations only after you provide them (or authorise a local run).

8. Status and approval

8.1 What I did

Identified and studied the reference dissertation, extracted its structure and exact Word formatting, inspected the CivicVoice source (models, routes, middleware, utilities, frontend routes, stack, PWA, deployment files), and prepared this plan. I did not write any dissertation content.

8.2 Verified

5 Mongoose models; 12 auth + 11 issue + 4 notification + 1 contact + 1 upload + 1 chat endpoints; JWT 7-day expiry; bcrypt hashing; OTP rate limits; 5-image/5 MB upload limits; Gemini priority/suggestions/solution flow; OSM tiles + Nominatim; PWA manifest; 33 Git commits.

8.3 Not verified / pending

The five discrepancies in Section 2; all student/institution details; screenshots; test evidence; performance data; final output format.

8.4 Awaiting your approval

Before I write, please answer the five decisions in Section 2 and provide the identity details in Section 6.2. If you approve the structure as-is and tell me to proceed with the implemented 4-state workflow and the actual (non-Framer/non-hot-toast) stack, I can begin drafting immediately.

— End of planning document —